

GENERAL SPECTRAL FLOORS FROM MARKED DEFORMATION PRESENTATIONS

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ABSTRACT. We assemble a finite algebraic construction of characteristic and minimal polynomial divisors preserved by a full normalized-tilt orbit. Chain operators supplied by a marked relation presentation generate an effective algebra on zeroth homology. Relation seeds then generate its smallest invariant source subspace. Together with compatible finite state, landing, and source maps, these data determine a static normalized sector and its largest invariant core. Without assuming invariance of the static sector, the common characteristic and minimal polynomial divisors of the same full tilt orbit are exactly those of that core. A finite spectral-avoidance argument proves both equalities. Global landing descent can be weakened to local descent for the stable image; when even local descent fails, a future-output quotient still retains well-defined dynamics. We give explicit closure tests, an observability-kernel formula for the stable core, and a finite closure algorithm. Exact rational models check the separate stages. The input bookkeeping is essential: the chain-operator marking and the state, landing, and source maps are supplied data, not consequences of a bare filtration. The contribution is their connection to the normalized-tilt quotient; the homological descent, invariant-subspace closure, and observability constructions are standard.

CONTENTS

1. Introduction and scope	1
2. Normalized tilts and spectral floors	2
3. Forced-sector extraction from finite landing data	5
4. Generating the normalized source	6
5. Syzygies and the constraint propagator	7
6. Reconstructing the effective equation algebra	8
7. Finite closure and assembly	9
8. Exact finite models and verification	9
9. Scope boundary and conclusion	11
References	11

1. INTRODUCTION AND SCOPE

Paper I [1] studies a fixed cubic contraction with a substantial recurrent spectrum. Paper II [2] shows that the normalized internal tilt orbit of its seven-dimensional envelope retains only the divisor $S+2$. This paper formulates the finite mechanism behind that distinction and connects it to a marked relation presentation.

Write $P_\bullet$ for a bounded complex of finite-dimensional vector spaces, concentrated in nonnegative homological degrees, and let $\mathfrak{G}_{\text{mark}}$ be a finite family of specified chain endomorphisms. A preserved finite filtration may also be supplied; it is needed only for the nilpotent-ideal

observation in Section 7. Let $R_0 : W_0 \rightarrow H_0(P_\bullet)$ be a seed map. These data determine an effective algebra and its generated seed module. To extract a spectral floor one also supplies the finite state and observation data:

$$\iota : H_0(P_\bullet) \hookrightarrow F, \quad T \in \text{End}(M), \quad \Lambda : M \rightarrow R, \quad D : R \rightarrow F.$$

Thus the full input is the marked package

$$\mathcal{D} = (P_\bullet, \mathfrak{G}_{\text{mark}}, R_0; \iota, F, M, T, \Lambda, D).$$

The stages, with their input dependencies understood, are

$$\begin{aligned} (1) \quad & (P_\bullet, \mathfrak{G}_{\text{mark}}, R_0) \mapsto \mathcal{A}_{\text{eq}}^{\text{eff}} \mapsto N_{\text{eq}} = \mathcal{A}_{\text{eq}}^{\text{eff}} \text{ im } R_0, \\ & (N_{\text{eq}}; \iota, D, \Lambda, T) \mapsto \Gamma_{\text{eq}} \mapsto (K_\infty, A_\Lambda|_{K_\infty}) \\ & \mapsto (\chi_{\text{forced}}, \mu_{\text{forced}}). \end{aligned}$$

Neither D, Λ, T nor the marked operator family is reconstructed from a bare complex. In particular, *removing external inputs* in the historical v0.57–v0.61 notes means replacing separately declared algebras and invariant subspaces by their generated versions, relative to this supplied data.

Homological descent and the stable-kernel computation are standard [3, 4, 5]. The operator family $A + \tau H$ is classical output injection, dual to state feedback. Section 2.1 makes the direct comparison with the pole-shifting theorem [6, Section 5.1, Theorem 13]. The contribution here is the marked-source assembly that produces this unchanged full orbit, not a new pole-assignment theorem.

2. NORMALIZED TILTS AND SPECTRAL FLOORS

Let $\mathbb{k}$ be an infinite field, let R and F be finite-dimensional $\mathbb{k}$ -vector spaces, and let

$$D : R \longrightarrow F, \quad N \subseteq F, \quad q : F \longrightarrow F/N$$

be linear data. Define the normalized kernel

$$(2) \quad K := D^{-1}(N), \quad A \in \text{End}(R).$$

The allowed normalized tilts are parametrized by $\tau \in \text{Hom}(\text{im}(qD), R)$, with

$$(3) \quad A_\tau := A + \tau qD.$$

The perturbation is invisible on K , so every representative has the same restriction to K .

Lemma 2.1 (Exact variation space). *The set of all perturbations obtained from normalized tilts is*

$$(4) \quad \{\tau qD : \tau \in \text{Hom}(\text{im}(qD), R)\} = \{T \in \text{End}(R) : T|_K = 0\}.$$

Proof. Every τqD vanishes on K . Conversely, if $T|_K = 0$, define $\tau(qDr) := Tr$. If $qDr = qDr'$, then $r - r' \in K$, and hence $T(r - r') = 0$. Thus τ is well-defined and $T = \tau qD$. $\square$

Theorem 2.2 (Normalized-tilt spectral floor). *Assume $A(K) \subseteq K$. With the characteristic and minimal polynomials of the zero-dimensional operator both defined as 1, one has*

$$\mathcal{O}_{\text{tilt}}(A) = \{B \in \text{End}(R) : B|_K = A|_K\},$$

and

$$(5) \quad \gcd_{\tau} \chi_{A_\tau} = \chi_{A|_K},$$

$$(6) \quad \gcd_{\tau} \mu_{A_\tau} = \mu_{A|_K}.$$

The gcd is monic and is taken over the full tilt family in (3).

Proof. The orbit formula follows from the exact variation lemma. Choose $R = K \oplus X$. The invariance assumption and the orbit formula give precisely the matrices

$$\begin{pmatrix} A|_K & C \\ 0 & B_X \end{pmatrix}, \quad C \in \text{Hom}(X, K), \quad B_X \in \text{End}(X).$$

Restriction implies that $\chi_{A|_K}$ and $\mu_{A|_K}$ divide the corresponding polynomial of every representative. If $X = 0$, the orbit is a singleton.

Otherwise choose two distinct scalars $c, d \in \mathbb{k}$ and take block-diagonal representatives $B_c = A|_K \oplus cI_X$ and $B_d = A|_K \oplus dI_X$. Writing $r = \dim X$, their characteristic polynomials are

$$\chi_{B_c} = \chi_{A|_K}(S - c)^r, \quad \chi_{B_d} = \chi_{A|_K}(S - d)^r,$$

whose gcd is $\chi_{A|_K}$. For minimal polynomials,

$$\mu_{B_c} = \text{lcm}(\mu_{A|_K}, S - c), \quad \mu_{B_d} = \text{lcm}(\mu_{A|_K}, S - d).$$

Their gcd is $\mu_{A|_K}$, as is seen by comparing each irreducible-factor valuation. The universal gcd divides each of these two-representative gcds, so the lower bounds are equalities. The argument also covers $K = 0$. $\square$

Remark 2.3 (Cubic specialization). For Paper II's internal orbit, take $R = \mathcal{R}$, $A = -hD_\alpha|_{\mathcal{R}}$ and $H = q_F D_\alpha$. Then $\ker H = \mathbb{C}E$ is invariant with $A(E) = -2E$. Theorem 2.2 gives both floors equal to $S+2$, although the selected seven-dimensional operator has a larger characteristic polynomial. This specializes the tilt theorem; it does not supply a canonical marked complex for the cubic equations.

Corollary 2.4 (Jordan-depth interpretation). *Over a splitting field, if*

$$\chi_{A|_K}(S) = \prod_{\lambda} (S - \lambda)^{c_\lambda}, \quad \mu_{A|_K}(S) = \prod_{\lambda} (S - \lambda)^{j_\lambda},$$

then c_λ and j_λ are respectively the algebraic-multiplicity and Jordan-depth floors that survive every normalized tilt.

Remark 2.5. The proof uses the full normalized variation space in (4). If an application restricts the tilt parameters further, the gcd can be larger. The theorem therefore identifies the floor for the stated orbit, not for every conceivable deformation.

2.1. The same orbit without invariant normalization. The assumption $A(K) \subseteq K$ simplifies the answer but is not needed to compute the floor of the original full orbit. Put $H = qD$, $n = \dim R > 0$, and

$$U = \bigcap_{j=0}^{n-1} \ker(HA^j).$$

This is the usual unobservable subspace for the pair (A, H) [6, Section 6.2]. The nearest spectral comparison is Sontag's pole-shifting theorem [6, Section 5.1, Theorem 13, p. 186]: transposition converts $A + \tau H$ to state feedback, and the fixed uncontrollable factor becomes the characteristic polynomial on U . Thus the characteristic floor is a dual output-injection consequence of that classical result. The minimal-polynomial equality also uses a coprime block decomposition; we prove that step below instead of attributing the equality directly to pole assignment. Our spectral-avoidance lemma provides a self-contained proof requiring only the weaker ability to avoid a finite spectral set.

Lemma 2.6 (Avoiding a finite spectral set). *Let $B \in \text{End}(X)$, $C : X \rightarrow Y$, and suppose $\bigcap_{j \geq 0} \ker(CB^j) = 0$. Over an infinite field $\mathbb{k}$, for every nonzero polynomial $f \in \mathbb{k}[S]$ there is $L : Y \rightarrow X$ such that*

$$\gcd(\chi_{B+LC}, f) = 1.$$

Proof. Work first over an algebraic closure. For any scalar λ , the map $\lambda I - B$ is injective on $\ker C$: a vector in its kernel would be an eigenvector invisible under every CB^j . Choose $X = \ker C \oplus X_1$. On X_1 , the value of LC can be prescribed arbitrarily because $C|_{X_1}$ is injective. Choose these values so that the images of X_1 under $\lambda I - B - LC$ complement $(\lambda I - B)(\ker C)$. The resulting matrix is invertible. Thus $\det(\lambda I - B - LC)$, as a polynomial in the entries of L , is not identically zero for any root λ of f . Their finite product is nonzero, hence so is the resultant $\text{Res}_S(f, \chi_{B+LC})$, a polynomial over $\mathbb{k}$. A nonzero polynomial has a nonzero value at some point of an affine space over an infinite field. That value gives the required L . The zero-dimensional and constant- f cases are immediate. $\square$

Theorem 2.7 (Stable-core floor for the unchanged full orbit). *For arbitrary A , with no assumption that $K = \ker H$ is invariant, U is the largest A -invariant subspace contained in K . Every representative $A_\tau = A + \tau H$ has the same subspace U , and the same action there. For this original tilt family,*

$$\boxed{\gcd_{\tau} \chi_{A_\tau} = \chi_{A|_U}, \quad \gcd_{\tau} \mu_{A_\tau} = \mu_{A|_U}.$$

The gcds can each be witnessed by the same two representatives. If K is invariant, $U = K$, recovering Theorem 2.2.

Proof. Cayley–Hamilton identifies the finite intersection defining U with the intersection over all $j \geq 0$, proving its invariance and maximality. On any invariant subspace in $\ker H$, A_τ and A agree. Applying this observation in both directions proves that the maximal such subspace is the same throughout the orbit.

Choose $R = U \oplus X$. Then A and H have the forms

$$A = \begin{pmatrix} A_U & C_0 \\ 0 & B \end{pmatrix}, \quad H = (0, C).$$

The induced pair (B, C) on R/U is observable: an invisible quotient vector lifts to a vector in U , by the infinite-intersection formula. The quotient blocks of the original tilts range over all $B + LC$. Every characteristic and minimal polynomial therefore contains the respective polynomial of A_U .

By Lemma 2.6, choose a first quotient block with characteristic polynomial p_1 coprime to χ_{A_U} , then a second with characteristic polynomial p_2 coprime to $\chi_{A_U} p_1$. Lift these choices to two original tilt parameters; their upper off-diagonal blocks need not vanish. Their characteristic polynomials are $\chi_{A_U} p_1$ and $\chi_{A_U} p_2$, whose gcd is χ_{A_U} .

For minimal polynomials the coprime spectra also remove the extension: if a block-triangular operator has diagonal blocks with coprime characteristic polynomials r, p , then $r(A)p(A) = 0$, and the Bezout identity decomposes its space as $\ker r(A) \oplus \ker p(A)$. The first summand is U : its quotient image is zero since r is invertible on the quotient, and $r(A)$ vanishes on U . The other summand identifies with the quotient. Hence for the two representatives

$$\mu_{A_{\tau_i}} = \mu_{A_U} \mu_{B+L_i C}, \quad i = 1, 2.$$

The two complementary minimal polynomials divide the coprime p_1, p_2 ; their gcd is one. This proves the minimal-polynomial equality. If $U = R$, the orbit is a singleton; if $U = 0$, the same argument gives both floors equal to one. The case $R = 0$ uses the same convention. $\square$

Remark 2.8. The theorem does not enlarge the allowed tilt family to all perturbations vanishing only on U . The constraints on the larger, possibly noninvariant K remain in place. The avoidance lemma proves the upper bound within that unchanged family.

3. FORCED-SECTOR EXTRACTION FROM FINITE LANDING DATA

Let $M, T, \Lambda, D, F, N, q$ be finite linear data as above, and put

$$R_\Lambda = \text{im } \Lambda, \quad \Gamma = qD\Lambda, \quad K_\Lambda = R_\Lambda \cap D^{-1}(N).$$

The distinction between R and R_Λ matters: the landing calculation describes the visible space R_Λ , unless Λ is surjective onto R .

Proposition 3.1 (Static extraction). *There are natural vector-space identifications*

$$(7) \quad K_\Lambda = \Lambda(\ker \Gamma) \simeq \ker \Gamma / \ker \Lambda, \quad \dim K_\Lambda = \text{rank } \Lambda - \text{rank } \Gamma.$$

Proof. One has $\Gamma m = 0$ exactly when $\Lambda m \in D^{-1}(N)$. The kernel of $\Lambda|_{\ker \Gamma}$ is $\ker \Lambda$. Apply the first isomorphism theorem and rank-nullity. $\square$

Proposition 3.2 (Landing descent and normalized closure). *An operator $A_\Lambda \in \text{End}(R_\Lambda)$ satisfying $A_\Lambda \Lambda = \Lambda T$ exists uniquely if and only if*

$$T \ker \Lambda \subseteq \ker \Lambda, \quad \text{equivalently} \quad \text{rank} \begin{bmatrix} \Lambda \\ \Lambda T \end{bmatrix} = \text{rank } \Lambda.$$

Assuming this descent condition, K_Λ is A_Λ -invariant if and only if

$$T \ker \Gamma \subseteq \ker \Gamma, \quad \text{equivalently} \quad \text{rank} \begin{bmatrix} \Gamma \\ \Gamma T \end{bmatrix} = \text{rank } \Gamma.$$

When both hold, $A_\Lambda|_{K_\Lambda}$ is conjugate to the operator induced by T on $\ker \Gamma / \ker \Lambda$.

Proof. The formula $A_\Lambda(\Lambda m) = \Lambda T m$ is well-defined exactly under the first kernel condition. Given it, for $m \in \ker \Gamma$ one has $A_\Lambda \Lambda m \in K_\Lambda$ exactly when $\Gamma T m = 0$. Finally $\ker L \subseteq \ker LT$ is equivalent to equality of the displayed stacked rank with $\text{rank } L$. $\square$

Proposition 3.3 (Stable core). *Let $m = \dim M > 0$ and*

$$\mathcal{O} = \begin{bmatrix} \Gamma \\ \Gamma T \\ \vdots \\ \Gamma T^{m-1} \end{bmatrix}, \quad M_\infty = \ker \mathcal{O}.$$

Then $M_\infty = \bigcap_{j \geq 0} \ker(\Gamma T^j)$ is the largest T -invariant subspace contained in $\ker \Gamma$. In general its landed image satisfies

$$\Lambda(M_\infty) \simeq M_\infty / (M_\infty \cap \ker \Lambda).$$

If the landing descent condition of Proposition 3.2 holds, then $\ker \Lambda \subseteq M_\infty$ and

$$K_\infty := \Lambda(M_\infty) \simeq M_\infty / \ker \Lambda, \quad \dim K_\infty = \text{rank } \Lambda - \text{rank } \mathcal{O}.$$

It is the largest A_Λ -invariant subspace of K_Λ .

Proof. Cayley–Hamilton reduces every power T^j , $j \geq m$, to the first m powers, proving the infinite-intersection equality and invariance. Any invariant subspace in $\ker \Gamma$ is in this intersection. The general quotient formula is rank-nullity for $\Lambda|_{M_\infty}$. Under landing descent, $\Lambda T^j n = 0$ for $n \in \ker \Lambda$, so $n \in M_\infty$. If $L \subseteq K_\Lambda$ is A_Λ -invariant, then $\Lambda^{-1}(L)$ is T -invariant and contained in $\ker \Gamma$, hence is contained in M_∞ . $\square$

This is the classical unobservable-subspace construction [4, 5], used here for the normalization output Γ . For $M = 0$ all spaces and ranks are zero. Under landing descent, Theorem 2.7 identifies the floor of the original landed tilt orbit with the operator on K_∞ , even if normalized closure fails. There is no need to replace the tilt orbit by a less constrained one.

Proposition 3.4 (Local descent and the future-output quotient). *Even without global landing descent, set $W = M_\infty$ and $J = W \cap \ker \Lambda$. A well-defined operator on ΛW with $A_W \Lambda w = \Lambda T w$ exists if and only if $TJ \subseteq J$. When it exists, it is conjugate to the action on W/J .*

Without that local condition, let

$$J_{\text{fut}} = \bigcap_{j=0}^{m-1} \ker(\Lambda T^j).$$

Then $J_{\text{fut}} \subseteq W$ is invariant, and W/J_{fut} always has an induced operator. It identifies with the image of the finite future-output map

$$w \mapsto (\Lambda w, \Lambda T w, \dots, \Lambda T^{m-1} w), \quad w \in W.$$

It is the smallest quotient of W preserving every future landing output: any quotient through which all maps $\Lambda T^j|_W$ factor has its kernel contained in J_{fut} .

Proof. W is T -invariant. Thus the formula on ΛW is well-defined exactly when the kernel J of $\Lambda|_W$ is T -invariant. Cayley–Hamilton makes J_{fut} invariant and replaces its finite intersection by the infinite one. Since $\Gamma = qD\Lambda$, invisibility under every ΛT^j implies invisibility under every ΓT^j ; hence $J_{\text{fut}} \subseteq W$. The first isomorphism theorem proves the finite-output identification. The final claim is exactly the intersection-of-kernels criterion for factorization. $\square$

Remark 3.5. The future-output quotient retains history, not just the static point Λw . It can have larger dimension than ΛW . It does not invent an operator on the original static landing space when descent fails, nor does it remove Λ or T from the supplied data.

Corollary 3.6 (Cyclic polynomial module). *Let $M = \mathbb{k}[S]/(c)$, with T acting by multiplication by S . Suppose both kernels are submodules and*

$$\ker \Lambda = (d_\Lambda)/(c), \quad \ker \Gamma = (d_\Gamma)/(c),$$

where the displayed divisors of c are monic. Then $d_\Gamma \mid d_\Lambda$, and

$$K_\Lambda \simeq \mathbb{k}[S]/(d_\Lambda/d_\Gamma).$$

The characteristic and minimal polynomials of the induced cyclic operator are both d_Λ/d_Γ , with the zero-module convention equal to 1.

Proof. Kernel inclusion reverses divisibility of the generators. Multiplication by d_Γ identifies $\mathbb{k}[S]/(d_\Lambda/d_\Gamma)$ with $(d_\Gamma)/(d_\Lambda)$, which is the required kernel quotient. $\square$

4. GENERATING THE NORMALIZED SOURCE

The normalized source should not be postulated when it can be generated from relation data. Let W_{rel} be a finite relation-seed space, let

$$R_0 : W_{\text{rel}} \longrightarrow F$$

be the seed map, and let $C_1, \dots, C_s \in \text{End}(F)$ be source-side constraint propagators. Starting from $N^{(0)} = \text{im } R_0$, define

$$(8) \quad N^{(r+1)} = N^{(r)} + \sum_{i=1}^s C_i N^{(r)}, \quad N_{\text{eq}} := \bigcup_{r \geq 0} N^{(r)}.$$

Because F is finite-dimensional, the increasing sequence stabilizes.

Proposition 4.1 (Minimal invariant source). *The stable space in (8) is*

$$(9) \quad N_{\text{eq}} = \mathbb{k}\langle C_1, \dots, C_s \rangle \text{im } R_0,$$

and it is the unique smallest subspace of F that contains $\text{im } R_0$ and is invariant under every C_i .

Proof. Induction gives $N^{(r)}$ equal to the span of all words in the C_i of length at most r applied to $\text{im } R_0$. The stable union is therefore the right-hand side of (9). Any invariant subspace containing the seed contains every such word, proving minimality. $\square$

Proposition 4.2 (Functoriality). *Suppose $\Phi_F : F \rightarrow F'$ and $\Phi_W : W_{\text{rel}} \rightarrow W'_{\text{rel}}$ satisfy*

$$\Phi_F R_0 = R'_0 \Phi_W, \quad \Phi_F C_i = C'_i \Phi_F.$$

Then $\Phi_F(N_{\text{eq}}) \subseteq N'_{\text{eq}}$. If Φ_F and Φ_W are isomorphisms intertwining all data, it carries N_{eq} isomorphically to N'_{eq} . More generally, when Φ_F is invertible, equality holds if and only if

$$\text{im } R'_0 \subseteq \Phi_F(N_{\text{eq}}).$$

In particular, mapping the seed image onto the target seed image is sufficient, but not necessary: it suffices to generate every target seed.

Proof. The intertwining identities send every operator word applied to a seed to the corresponding target word. This proves inclusion. With an invertible ambient map, $\Phi_F(N_{\text{eq}})$ is invariant under every target propagator. If it contains the target seed image, minimality gives the reverse inclusion. Conversely equality necessarily contains that image. $\square$

In the source-generation step the ambient space can be the relation module itself. If it embeds in a larger source space by ι , use $N = \iota(N_{\text{eq}})$ there. The landed forced sector is then

$$\Gamma_{\text{eq}} = q_{\text{eq}} D\Lambda, \quad K_{\Lambda} \simeq \ker \Gamma_{\text{eq}} / \ker \Lambda,$$

with the closure conditions of Proposition 3.2. No choice of propagators outside the embedded relation module is needed.

5. SYZYGIES AND THE CONSTRAINT PROPAGATOR

The source propagators themselves must come from an equation action. Let $\mathcal{A}_{\text{eq}}$ temporarily denote a unital algebra and let F_{rel} be a finite-dimensional left module. Let

$$P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} F_{\text{rel}} \longrightarrow 0$$

be an exact presentation in left $\mathcal{A}_{\text{eq}}$ -modules. Equivalently, $F_{\text{rel}} = H_0(P_{\bullet})$ for the two-term complex.

Proposition 5.1 (Syzygy descent). *If $\mathcal{A}_{\text{eq}}$ acts on the presentation complex by chain maps, it induces a representation $\bar{\rho} : \mathcal{A}_{\text{eq}} \rightarrow \text{End}(H_0(P_{\bullet}))$. Let $\bar{\varepsilon} : H_0(P_{\bullet}) \xrightarrow{\sim} F_{\text{rel}}$ be the presentation isomorphism. The induced source constraint propagator is*

$$(10) \quad C(a) = \bar{\varepsilon} \bar{\rho}(a) \bar{\varepsilon}^{-1} \quad \text{on the identified relation image,}$$

and the effective constraint algebra is

$$(11) \quad \mathcal{C}_{\text{syz}} = \text{im}(C : \mathcal{A}_{\text{eq}} \rightarrow \text{End}(F_{\text{rel}})) \simeq \mathcal{A}_{\text{eq}} / \text{Ann}(F_{\text{rel}}).$$

Proof. The chain-map condition preserves $\text{im } d_1$, so the action on P_0 descends to $P_0 / \text{im } d_1 = H_0(P_{\bullet})$. Transport along the presentation isomorphism gives (10). The kernel of the resulting representation is exactly the annihilator of the relation module, which proves (11). $\square$

Changing generators or replacing the presentation by an isomorphic presentation complex changes matrices but not the induced representation on H_0 . An ambient extension of $C(a)$ outside the relation image is noncanonical and has no role in N_{eq} . This is the precise syzygy-to-propagator bridge needed by the source-closure step.

6. RECONSTRUCTING THE EFFECTIVE EQUATION ALGEBRA

The temporary algebra $\mathcal{A}_{\text{eq}}$ in Section 5 is now removed from the external input. Let $P_\bullet$ be a bounded finite-dimensional relation complex and let

$$\mathfrak{G}_{\text{mark}} = \{G_1, \dots, G_t\}$$

be the specified marking. No acyclicity in positive homological degrees is required: only the induced action on $H_0(P_\bullet)$ is used. Assume each operator is a chain map:

$$dG_j = G_j d.$$

Write $H_0 = H_0(P_\bullet)$. Each G_j descends to $\bar{G}_j \in \text{End}(H_0)$.

Definition 6.1 (Effective equation-operator algebra). The effective algebra of the marked presentation is

$$(12) \quad \mathcal{A}_{\text{eq}}^{\text{eff}} := \mathbb{k}\langle \bar{G} : G \in \mathfrak{G}_{\text{mark}} \rangle \subseteq \text{End}(H_0).$$

Equivalently, if $\mathbb{k}\langle X_G : G \in \mathfrak{G}_{\text{mark}} \rangle$ is the free unital algebra, evaluation gives

$$(13) \quad \text{im}(\text{ev}_P) \simeq \mathbb{k}\langle X_G : G \in \mathfrak{G}_{\text{mark}} \rangle / \ker(\text{ev}_P).$$

Proposition 6.2 (Minimality and comparison). *The algebra $\mathcal{A}_{\text{eq}}^{\text{eff}}$ is the smallest unital subalgebra of $\text{End}(H_0)$ containing the descended operators. Suppose the two marked families are indexed in correspondence and a chain map $\Phi : P_\bullet \rightarrow P'_\bullet$ satisfies*

$$\Phi G - G' \Phi = d' H_G + H_G d$$

for degree-one chain homotopies $H_G : P_\bullet \rightarrow P'_{\bullet+1}$. Then $\bar{\Phi} \bar{G} = \bar{G}' \bar{\Phi}$ on H_0 . If $\bar{\Phi}$ is an isomorphism, conjugation by $\bar{\Phi}$ identifies the two effective algebras.

Proof. The chain-map condition preserves boundaries, so descent is well-defined. The homotopy equation induces equality on homology [3]. Intertwining the generators intertwines every evaluated word; an invertible $\bar{\Phi}$ gives the stated conjugacy. Formula (13) is the first isomorphism theorem for the evaluation representation. $\square$

A noninvertible comparison still intertwines evaluated words; an isomorphism of effective algebras does not follow without the extra hypothesis. Likewise, adding a contractible direct summand has no effect when the descended marked actions are identified, even if the raw actions on the acyclic summand differ. Transporting through the presentation identification gives

$$\mathcal{C}_{\text{syz}} \simeq \mathcal{A}_{\text{eq}}^{\text{eff}}, \quad N_{\text{eq}} = \mathcal{A}_{\text{eq}}^{\text{eff}} \text{ im } R_0.$$

Here the first equality is a canonical identification of represented algebras, rather than a literal equality of matrices in unrelated bases.

7. FINITE CLOSURE AND ASSEMBLY

Put $h = \dim H_0$. Starting with $V_0 = \mathbb{k}I$, form

$$V_{j+1} = V_j + \sum_{G \in \mathfrak{G}_{\text{mark}}} \bar{G}V_j.$$

By induction V_j is the span of words of length at most j . Once $V_{j+1} = V_j$, all longer words lie in the same space, so it equals $\mathcal{A}_{\text{eq}}^{\text{eff}}$. There are at most h^2 strict dimension increases; if $h > 0$ the initial identity improves this bound to $h^2 - 1$. Source closure similarly terminates after at most $\dim F_{\text{rel}}$ strict increases. No filtration hypothesis is needed for this termination argument.

If $H_0 = F^0 H_0 \supseteq \cdots \supseteq F^L H_0 = 0$ is a finite descending filtration preserved by $\mathcal{A}_{\text{eq}}^{\text{eff}}$, define

$$J = \{a \in \mathcal{A}_{\text{eq}}^{\text{eff}} : a(F^p H_0) \subseteq F^{p+1} H_0 \text{ for every } p\}.$$

Then J is a two-sided ideal and $J^L = 0$: products add filtration shifts. Thus the ideal generated by any positive-degree marked operators is nilpotent. This conclusion requires filtration preservation by the entire represented algebra.

Theorem 7.1 (Stable assembly for finite marked data). *Let $\mathcal{D}$ be the data of Section 1. Define $N_{\text{eq}} = \mathcal{A}_{\text{eq}}^{\text{eff}} \text{ im } R_0 \subseteq H_0(P_\bullet)$, let $q_{\text{eq}} : F \rightarrow F/\iota(N_{\text{eq}})$, and put $\Gamma_{\text{eq}} = q_{\text{eq}} D\Lambda$. Assume only the landing descent condition*

$$T \ker \Lambda \subseteq \ker \Lambda.$$

For $m = \dim M > 0$, set

$$W = \bigcap_{j=0}^{m-1} \ker(\Gamma_{\text{eq}} T^j).$$

For the full original tilt family on R_Λ , with source map $D|_{R_\Lambda}$ and normalization $\iota(N_{\text{eq}})$, the forced characteristic and minimal polynomials are exactly those of the T -action on

$$W/\ker \Lambda.$$

If $T \ker \Gamma_{\text{eq}} \subseteq \ker \Gamma_{\text{eq}}$, this specializes to the static quotient $\ker \Gamma_{\text{eq}}/\ker \Lambda$. For $M = 0$, both polynomials are one.

Proof. Source closure computes N_{eq} . Landing descent defines A_Λ . The stable-core proposition gives $\ker \Lambda \subseteq W$ and identifies ΛW with the largest A_Λ -invariant subspace of K_Λ . Theorem 2.7, applied to the original family on R_Λ , gives the result. Under normalized closure, $W = \ker \Gamma_{\text{eq}}$, recovering the earlier static formula. $\square$

The output is transported by isomorphisms of the full marked data: in addition to Proposition 6.2, the seed images and the maps ι, D, Λ, T must be intertwined. Naturality of the effective algebra alone is insufficient to compare two different landing systems.

8. EXACT FINITE MODELS AND VERIFICATION

8.1. Equation-operator reconstruction model. For this model take $\mathbb{k} = \mathbb{Q}$, $H_0 = \mathbb{Q}^4$, and $Se_i = e_{i+1}$ for $i < 4$, $Se_4 = 0$, so S is the lower nilpotent shift, and let

$$W = \text{diag}(0, 1, 2, 3).$$

Then

$$S^4 = 0, \quad [W, S] = S.$$

The algebra generated by S and W is the lower-triangular matrix algebra in this basis, with dimension 10. Indeed the distinct diagonal weights give all four diagonal matrix units by

polynomial interpolation in W , and $E_{ii}S^{i-j}E_{jj} = E_{ij}$ for $i \geq j$. The shift-only algebra $\mathbb{k}[S]$ has dimension 4. A change of basis preserves these dimensions and relations after conjugation. Adding a contractible summand changes raw chain matrices but leaves the induced H_0 action unchanged. These checks verify precisely the claims that matter for v0.61; they do not assert that arbitrary chain complexes carry such operators canonically.

8.2. Earlier stages of the pipeline. The exact finite checks in the v0.57–v0.61 records [7] give the following values:

TABLE 1. Exact checks for the separate finite stages.

Version	finite output	role
v0.57	$\chi_{\text{forced}} = \mu_{\text{forced}} = (S+2)^2(S-3)$, tilt dimension 18	floor theorem
v0.58	forced-sector dimension 3; cyclic floor $(S+2)^2$	quotient extraction
v0.59	ranks 2, 4, 5, 5; $\dim N_{\text{eq}} = 5$	source closure
v0.60	syzygy rank 4; quotient dimension 4; λ^4	propagator bridge
v0.61	$\dim \mathcal{A}_{\text{eq}}^{\text{eff}} = 10$, shift-only dimension 4	operator reconstruction

The five historical-stage verifier scripts, together with the bundled recovery models, are finite symbolic or exact matrix checks. Their PASS outputs are evidence for the listed models and identities. They are not substitutes for the hypotheses in Theorems 2.2 and the propositions above.

8.3. A noninvariant normalization with an exact floor. Take

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix}, \quad H = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}.$$

Then $K = \ker H = \text{span}(e_1, e_2)$ is not invariant, but $U = \text{span}(e_2)$. The original orbit allows changes only to the third column. The parameters $\tau_0 = (0, 0, -3)^\top$ and $\tau_1 = (-1, 0, -3)^\top$ give

$$\chi_{A+\tau_0 H} = (S-2)S^2, \quad \chi_{A+\tau_1 H} = (S-2)(S^2+1).$$

These two matrices are cyclic, so their minimal polynomials are the same as their characteristic polynomials. Both gcfs are exactly $S-2$. The supplement also tests noncyclic Jordan cores, rational changes of basis, multiple outputs, local descent without global descent, and generated-seed coverage without surjectivity onto the target seed image.

For an end-to-end marked realization, take $P_1 = \mathbb{Q}$, $P_0 = \mathbb{Q}^3$, $d_1(z) = (0, 0, z)$, and mark the chain map $G_0 = \text{diag}(1, 2, 3)$, $G_1 = 3$. Then $H_0(P_\bullet) = \mathbb{Q}^2 = F$, with the identity embedding and descended operator $\bar{G} = \text{diag}(1, 2)$. The seed $R_0(s) = (s, 0)$ generates $N = \mathbb{Q}(1, 0)$, while $\mathcal{A}_{\text{eq}}^{\text{eff}}$ is the two-dimensional diagonal algebra. With $M = R = \mathbb{Q}^3$, $T = A$, $\Lambda = I$, and

$$D(x_1, x_2, x_3) = (0, x_3),$$

one has $qD\Lambda = H$. Thus every arrow of the stable assembly is realized in one exact finite model, including a failure of normalized closure. This is an explicit marked model, not a claim that the same inputs arise canonically in every inverse-Leibniz presentation.

9. SCOPE BOUNDARY AND CONCLUSION

A bare complex and its filtration cannot select an arbitrary distinguished endomorphism family. For example, take $P_0 = \mathbb{Q}^4$, all other $P_i = 0$, and the descending coordinate flag preserved by lower triangular matrices. The marks $\{S\}$ and $\{S, W\}$ above are both allowed on that same filtered complex, but their generated algebras have dimensions 4 and 10. The unmarked input does not determine which marked algebra was intended. This does not exclude intrinsically distinguished scalar maps such as the identity, or a more restrictive geometric construction that supplies additional operators.

Within the specified marked category, the sequence v0.57–v0.61 gives a finite assembly procedure: equation action on homology, invariant source closure, kernel-quotient extraction, and the normalized-tilt polynomial floor. Its hypotheses explicitly retain the chain operations and the state, landing, and source data. The exact models establish reproducible finite checks of the stages. They do not prove that all of those inputs arise canonically from an arbitrary inverse-Leibniz deformation problem.

The next separate question is whether a chosen deformation presentation supplies its operator family by a natural construction. Paper II [2] already shows why an ordinary quasi-isomorphism class can lose boundary information. The current theorem therefore stops at the finite marked data of Theorem 7.1.

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